Experiment 22 — Smart Waste Management System

Aim:

To develop a web-based waste management system that enables administrators to manage garbage bins and drivers, the public to register complaints, and drivers to view assigned routes and update garbage collection status.

Algorithm:

1. Initialize the application and display the live waste management dashboard.
2. Allow the administrator to add, update, and delete garbage bins, manage drivers, and assign collection routes.
3. Allow the public to submit complaints and track their status.
4. Allow drivers to view assigned collection tasks, select routes, and update garbage load and collection status.
5. Display bin conditions, collection progress, complaints, and waste statistics using interactive charts and a map.
6. Refresh dashboard information periodically and save demonstration data in browser storage.
7. Generate waste collection reports and display route information for efficient collection.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1">

<title>EcoTrack | Smart Waste Management</title>


<link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css">

<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>
:root{--bg:#f3f6f2;--green:#176b4d;--mint:#dff5e8;--dark:#163c31;
--text:#20332c;--muted:#728079;--white:#fff;--orange:#f5a623;--red:#d95050}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--text);font:14px Arial,sans-serif}
header{background:var(--dark);color:white;padding:17px 4%;display:flex;
align-items:center;justify-content:space-between;gap:15px;flex-wrap:wrap}
.brand{font-size:23px;font-weight:bold}.brand span{color:#6ce0a5}
header small{color:#c7e5d5}
nav{display:flex;gap:7px;flex-wrap:wrap}
button{border:0;padding:10px 13px;border-radius:8px;background:var(--green);
color:white;cursor:pointer;font-weight:bold}
button:hover{filter:brightness(1.12)}
button.secondary{background:#e7eee9;color:var(--dark)}
button.danger{background:var(--red)}
button.warn{background:#a96c09}
input,select,textarea{width:100%;padding:11px;margin:5px 0 12px;
border:1px solid #d9e3dc;border-radius:8px;font:inherit;background:white}
label{font-weight:bold;font-size:12px}
.layout{max-width:1450px;margin:auto;padding:23px}
.hero{min-height:235px;padding:30px;border-radius:18px;color:white;
background:linear-gradient(90deg,#102f25eF,#176b4dc7),
url('https://images.unsplash.com/photo-1532996122724-
e3c354a0b15b?auto=format&fit=crop&w=1600&q=85') center/cover;
display:flex;justify-content:space-between;align-items:center;gap:20px}
.hero h1{font-size:clamp(27px,4vw,43px);margin:8px 0}
```

```
.hero p{max-width:570px;line-height:1.7;color:#e0f2e8}
.eyebrow{letter-spacing:2px;font-size:11px;font-weight:bold;color:#8af0b6}
.pill{display:inline-block;padding:7px 10px;border-radius:20px;background:#e0f4e8;
color:#176b4d;font-size:11px;font-weight:bold}
.stats{display:grid;grid-template-
columns:repeat(4,minmax(0,1fr));gap:14px;margin:20px 0}
.stat,.panel{background:white;padding:19px;border-radius:14px;box-shadow:0 3px
15px #173d2a0a}
.stat .number{font-size:30px;font-weight:bold;margin:10px 0}
.muted{color:var(--muted);font-size:12px}
.grid{display:grid;grid-template-columns:1.3fr 1fr;gap:17px;margin-bottom:18px}
.panel h2,.panel h3{margin-top:0}
.panel h2{font-size:19px}
#map{height:340px;border-radius:12px;z-index:1}
.chartbox{height:280px}
.tabs{display:flex;gap:8px;flex-wrap:wrap;margin:22px 0}
.tabs button.active{background:var(--dark);outline:2px solid #8ae2b5}
.page{display:none}.page.active{display:block}
.formgrid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:8px}
.bin-grid{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:13px}
.bin-card{background:white;border-radius:13px;overflow:hidden;border:1px solid
#e2eae4}
.bin-card img{width:100%;height:115px;object-fit:cover}
.bin-info{padding:13px}.bin-info h3{margin:0 0 7px}
.progress{height:8px;border-
radius:10px;background:#e7eee9;overflow:hidden;margin:10px 0}
.progress div{height:100%;background:#36a873;transition:width .5s}
.tablewrap{overflow-x:auto}
table{width:100%;border-collapse:collapse;min-width:620px}
```

th,td{text-align:left;padding:12px 10px;border-bottom:1px solid #edf0ed}

th{font-size:11px;text-transform:uppercase;color:var(--muted)}

td{font-size:13px}

.tag{display:inline-block;padding:5px 8px;border-radius:7px;background:#e1f5e8;color:#176b4d;font-size:11px}

.tag.warn{background:#fff0d7;color:#96600a}

.tag.red{background:#ffe4e4;color:#ae3030}

.notice{padding:12px;background:#e7f6eb;border-radius:9px;margin-bottom:12px}

.empty{padding:24px;text-align:center;color:var(--muted)}

footer{background:var(--dark);color:#cfe5d9;padding:22px 4%;text-align:center;margin-top:25px}

#toast{display:none;position:fixed;bottom:20px;right:20px;z-index:5000;

padding:15px 20px;background:#163c31;color:white;border-radius:10px;

box-shadow:0 5px 20px #0002;max-width:320px}

.live-dot{display:inline-block;width:8px;height:8px;border-radius:50%;background:#4de09a;margin-right:6px}

@media(max-width:900px){.stats{grid-template-columns:repeat(2,1fr)}

.grid{grid-template-columns:1fr}.bin-grid{grid-template-columns:repeat(2,1fr)}}

@media(max-width:540px){.layout{padding:12px}.stats{gap:8px}

.stat{padding:12px}.stat .number{font-size:24px}.bin-grid{grid-template-columns:1fr}

.hero{padding:22px}.formgrid{grid-template-columns:1fr}header{padding:16px}

#map{height:290px}}

</style>

</head>

<body>

<header>

♻ EcoTrack

```
  <small>SMART WASTE MANAGEMENT NETWORK</small>
 </div>
 <nav>
  <button onclick="switchPage('dashboard')">Dashboard</button>
  <button onclick="switchPage('bins')">Garbage Bins</button>
  <button onclick="switchPage('public')">Public Complaints</button>
  <button onclick="switchPage('driver')">Driver Portal</button>
  <button onclick="switchPage('admin')">Admin Control</button>
 </nav>
</header>

<main class="layout">
<section class="hero">
 <div>
  <span class="eyebrow">CLEANER STREETS. SMARTER CITIES.</span>
  <h1>Waste management,<br>made intelligent.</h1>
  <p>Monitor bin levels, organize collection routes, manage public complaints
    and track daily collection activity from one interactive dashboard.</p>
  <button onclick="switchPage('bins')">Explore live bins ↗</button>
 </div>
 <div style="text-align:right;min-width:120px">
  <div style="font-size:58px">🌿</div>
  <span class="pill"><span class="live-dot"></span>Dashboard active</span>
 </div>
</section>

<div class="stats">
 <div class="stat"><div class="muted">TOTAL GARBAGE BINS</div>
```

```
  <div class="number" id="totalBins">0</div><span
class="pill">Registered</span></div>

 <div class="stat"><div class="muted">BINS NEEDING COLLECTION</div>

  <div class="number" id="urgentBins">0</div><span class="pill"
style="background:#fff0d7;color:#96600a">High fill level</span></div>

 <div class="stat"><div class="muted">WASTE COLLECTED TODAY</div>

  <div class="number"><span id="collected">0</span> <small>kg</small></div><span
class="pill">Collection log</span></div>

 <div class="stat"><div class="muted">OPEN COMPLAINTS</div>

  <div class="number" id="openComplaints">0</div><span class="pill"
style="background:#ffe4e4;color:#ae3030">Public reports</span></div>

</div>


<div class="tabs">

 <button class="active" data-page="dashboard"
onclick="switchPage('dashboard')">Overview</button>

 <button data-page="bins" onclick="switchPage('bins')">Bin Monitoring</button>

 <button data-page="public" onclick="switchPage('public')">Citizen Services</button>

 <button data-page="driver" onclick="switchPage('driver')">Driver Workspace</button>

 <button data-page="admin" onclick="switchPage('admin')">Administrator</button>

</div>


<!-- DASHBOARD -->

<section class="page active" id="dashboard">

 <div class="grid">

  <div class="panel">

   <h2>📍 Collection Network Map</h2>

   <p class="muted">Select a map marker to inspect a bin's current fill level.</p>

   <div id="map"></div>
```

```
    <p class="muted">Map markers use demonstration coordinates in Chennai.</p>
   </div>
   <div class="panel">
    <h2>📊 Waste Collection Analytics</h2>
    <p class="muted">Illustrative collection data for the last seven days.</p>
    <div class="chartbox"><canvas id="wasteChart"></canvas></div>
   </div>
  </div>
  <div class="grid">
   <div class="panel">
    <h2>⚠ Collection Priority</h2>
    <div id="priorityList"></div>
   </div>
   <div class="panel">
    <h2>🕒 Recent Activity</h2>
    <div id="activityList"></div>
    <p class="muted"><span class="live-dot"></span>Last refreshed: <span
id="lastUpdated">--</span></p>
   </div>
  </div>
</section>

<!-- BIN MONITORING -->
<section class="page" id="bins">
  <div class="panel">
   <h2>Garbage Bin Monitoring</h2>
   <div class="formgrid">
```

```
    <div><label>Search bin or area</label><input id="binSearch" placeholder="e.g. T Nagar" oninput="renderBins()"></div>
    <div><label>Fill level filter</label>
      <select id="binFilter" onchange="renderBins()">
        <option value="all">All bins</option><option value="urgent">Needs collection (75%+)</option>
        <option value="normal">Normal (below 75%)</option>
      </select>
    </div>
   </div>
  </div>
  <div class="bin-grid" id="binGrid"></div>
</section>

<!-- PUBLIC COMPLAINTS -->
<section class="page" id="public">
  <div class="grid">
   <div class="panel">
    <h2>📣 Report a Waste Issue</h2>
    <p class="muted">Report overflowing bins, missed collection or illegal dumping.</p>
    <form id="complaintForm">
      <label>Your name</label><input id="citizenName" required>
      <label>Contact email</label><input id="citizenEmail" type="email" required>
      <label>Issue category</label>
      <select id="issueType"><option>Overflowing bin</option><option>Missed collection</option>
        <option>Illegal dumping</option><option>Damaged bin</option></select>
```

```
      <label>Area / landmark</label><input id="issueArea" required placeholder="Street or locality">
      <label>Description</label><textarea id="issueDesc" required rows="3"></textarea>
      <button type="submit">Submit Complaint</button>
    </form>
   </div>
   <div class="panel">
    <h2>My Complaint Status</h2>
    <label>Complaint ID</label><input id="trackId" placeholder="e.g. CMP-123456">
    <button onclick="trackComplaint()">Track Complaint</button>
    <div id="trackResult" style="margin-top:15px"></div>
    <hr style="border:0;border-top:1px solid #eee;margin:25px 0">
    <h3>Service information</h3>
    <p>♻️ Separate recyclable and organic waste.</p>
    <p> 📍 Include a clear landmark in reports.</p>
    <p>🚛 Collection schedules are managed by the administrator.</p>
   </div>
  </div>
</section>

<!-- DRIVER WORKSPACE -->
<section class="page" id="driver">
 <div class="grid">
  <div class="panel">
   <h2>🚛 Driver Workspace</h2>
   <label>Driver name</label>
   <select id="driverName">
    <option>Arun Kumar</option><option>Priya S</option><option>Rahul M</option>
```

```
    </select>
    <p class="muted">Choose a bin to collect, update its load and record completion.</p>
    <label>Assigned bin</label><select id="driverBin"></select>
    <label>Collected waste (kg)</label><input type="number" id="loadKg" min="0" value="25">
    <button onclick="collectBin()">Complete Collection</button>
    <button class="secondary" onclick="buildRoute()">Suggest Route</button>
    <div id="routeInfo" class="notice" style="margin-top:15px">Select “Suggest Route” to prioritize bins.</div>
   </div>
   <div class="panel">
    <h2>Today's Collection Tasks</h2>
    <div id="driverTasks"></div>
   </div>
  </div>
</section>

<!-- ADMINISTRATOR -->
<section class="page" id="admin">
  <div class="panel">
   <h2>🔐 Administrator Control Center</h2>
   <p class="muted">Demo password: <b>admin123</b>. This browser-only login is not secure authentication.</p>
   <div class="formgrid">
    <div><label>Admin password</label><input type="password" id="adminPass"></div>
    <div style="align-self:end"><button onclick="adminLogin()">Unlock Admin Tools</button></div>
   </div>
```

```
  <div id="adminTools" style="display:none">
   <hr style="border:0;border-top:1px solid #eee;margin:20px 0">
   <h3>Create Garbage Bin</h3>
   <form id="binForm">
    <div class="formgrid">
     <div><label>Bin name</label><input id="newBinName" required placeholder="Central Market Bin"></div>
     <div><label>Area</label><input id="newBinArea" required placeholder="Chennai"></div>
     <div><label>Initial fill level (%)</label><input type="number" id="newBinFill" min="0" max="100" value="10" required></div>
     <div><label>Capacity (kg)</label><input type="number" id="newBinCapacity" min="1" value="100" required></div>
    </div>
    <button type="submit">+ Add Bin</button>
   </form>
   <h3 style="margin-top:25px">Manage Bins and Complaints</h3>
   <div class="tablewrap" id="adminBinTable"></div>
   <h3 style="margin-top:25px">Public Complaint Queue</h3>
   <div class="tablewrap" id="adminComplaintTable"></div>
   <div style="margin-top:18px;display:flex;gap:10px;flex-wrap:wrap">
    <button onclick="exportReport()">Download Waste Report CSV</button>
    <button class="secondary" onclick="resetDemo()">Reset Demo Data</button>
   </div>
  </div>
 </div>
</section>
</main>
```

<footer>EcoTrack • Smart City Waste Management • Educational Demonstration</footer>

<script>

const initialBins=[

{id:"BIN-101",name:"Marina Beach",area:"Marina Beach",fill:88,capacity:150,lat:13.0500,lng:80.2824,status:"Needs collection",collected:false},

{id:"BIN-102",name:"T Nagar Market",area:"T Nagar",fill:64,capacity:120,lat:13.0418,lng:80.2341,status:"Normal",collected:false},

{id:"BIN-103",name:"Anna Nagar West",area:"Anna Nagar",fill:34,capacity:100,lat:13.0850,lng:80.2101,status:"Normal",collected:false},

{id:"BIN-104",name:"Central Railway",area:"Egmore",fill:93,capacity:180,lat:13.0780,lng:80.2610,status:"Needs collection",collected:false},

{id:"BIN-105",name:"Adyar Junction",area:"Adyar",fill:49,capacity:110,lat:13.0067,lng:80.2206,status:"Normal",collected:false},

{id:"BIN-106",name:"Velachery Main",area:"Velachery",fill:78,capacity:130,lat:12.9815,lng:80.2180,status:"Needs collection",collected:false}

];

const initialComplaints=[];

function read(key,fallback){

try{return JSON.parse(localStorage.getItem(key))??fallback}catch(e){return fallback}

}

let bins=read("eco_bins",initialBins);

let complaints=read("eco_complaints",initialComplaints);

let activity=read("eco_activity",["System initialized","Collection dashboard opened"]);

let collectedKg=read("eco_collected",125);

```
let chart=null,map=null,markers={},adminUnlocked=false;


function persist(){
 localStorage.setItem("eco_bins",JSON.stringify(bins));
 localStorage.setItem("eco_complaints",JSON.stringify(complaints));
 localStorage.setItem("eco_activity",JSON.stringify(activity.slice(0,30)));
 localStorage.setItem("eco_collected",JSON.stringify(collectedKg));
}
function safe(s){
 return
String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","":"&quot;","'":"&#39
;"}[c]));
}
function toast(s){
 const t=document.getElementById("toast");t.textContent=s;t.style.display="block";

clearTimeout(window.toastTimer);window.toastTimer=setTimeout(()=>t.style.display="n
one",3200);
}
function log(s){
 activity.unshift(new Date().toLocaleTimeString()+" —
"+s);activity=activity.slice(0,30);persist();
 renderActivity();
}
function switchPage(id){
 document.querySelectorAll(".page").forEach(p=>p.classList.remove("active"));
 document.getElementById(id).classList.add("active");
 document.querySelectorAll("[data-
page]").forEach(b=>b.classList.toggle("active",b.dataset.page===id));
 if(id==="bins")renderBins();
```

```
 if(id==="driver")renderDriver();
 if(id==="admin"&&adminUnlocked)renderAdmin();
 if(id==="dashboard")setTimeout(()=>map&&map.invalidateSize(),100);
 window.scrollTo({top:0,behavior:"smooth"});
}
function updateStats(){
 document.getElementById("totalBins").textContent=bins.length;
 document.getElementById("urgentBins").textContent=bins.filter(b=>b.fill>=75).length;
 document.getElementById("collected").textContent=collectedKg.toLocaleString("en-IN");

document.getElementById("openComplaints").textContent=complaints.filter(c=>c.status!=="Resolved").length;
 document.getElementById("lastUpdated").textContent=new Date().toLocaleTimeString();
}
function renderBins(){
 const q=document.getElementById("binSearch").value.toLowerCase();
 const f=document.getElementById("binFilter").value;
 const list=bins.filter(b=>(b.name+" "+b.area+" "+b.id).toLowerCase().includes(q)&&
 (f==="all"||(f==="urgent"&&b.fill>=75)||(f==="normal"&&b.fill<75)));
 document.getElementById("binGrid").innerHTML=list.length?list.map(b=>`
 <article class="bin-card">
 <img src="https://images.unsplash.com/photo-1532996122724-e3c354a0b15b?auto=format&fit=crop&w=500&q=75" alt="Waste collection">
 <div class="bin-info"><span class="pill">${safe(b.id)}</span>
 <h3 style="margin-top:10px">${safe(b.name)}</h3><p class="muted">📍 ${safe(b.area)}</p>
 <div style="display:flex;justify-content:space-between"><b>Fill level</b><b>${b.fill}%</b></div>
```

```
 <div class="progress"><div
style="width:${b.fill}%;background:${b.fill>=85?'#d95050':b.fill>=75?'#f5a623':'#36a873
'}"></div></div>

 <p class="muted">Capacity: ${b.capacity} kg · ${safe(b.status)}</p>

 <button onclick="chooseDriverBin('${b.id}')">Assign for
collection</button></div></article>`).join(""):'<div class="empty">No bins match this
filter.</div>';

}

function initMap(){

 if(!window.L)return;

 map=L.map("map").setView([13.055,80.235],11);

 L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png",{

 attribution:'&copy; OpenStreetMap contributors',maxZoom:19

 }).addTo(map);

 renderMarkers();

}

function renderMarkers(){

 if(!map)return;

 Object.values(markers).forEach(m=>map.removeLayer(m));markers={};

 bins.forEach(b=>{

  const color=b.fill>=85?"red":b.fill>=75?"orange":"green";

  const marker=L.circleMarker([b.lat,b.lng],{

   radius:9,color,fillColor:color,fillOpacity:.8

  }).addTo(map);

  marker.bindPopup(`<b>${safe(b.name)}</b><br>${safe(b.id)}<br>Fill:
${b.fill}%<br>Status: ${safe(b.status)}`);

  markers[b.id]=marker;

 });

}
```

```
function renderPriority(){
 const urgent=[...bins].filter(b=>b.fill>=75).sort((a,b)=>b.fill-a.fill);
 document.getElementById("priorityList").innerHTML=urgent.length?urgent.map(b=>`
 <div style="display:flex;justify-content:space-between;gap:10px;padding:12px 0;border-bottom:1px solid #eee">
 <div><b>${safe(b.name)}</b><div class="muted">${safe(b.area)} · ${safe(b.id)}</div></div>
 <span class="tag ${b.fill>=85?'red':'warn'}">${b.fill}% full</span></div>`).join(""):'<div class="notice">All bins are below the collection threshold.</div>';
}
function renderActivity(){
 document.getElementById("activityList").innerHTML=activity.slice(0,6).map(a=>`
 <p style="border-bottom:1px solid #eee;padding-bottom:10px">• ${safe(a)}</p>`).join("");
}
function renderDriver(){
 document.getElementById("driverBin").innerHTML=bins.map(b=>
 `<option value="${safe(b.id)}">${safe(b.name)} — ${b.fill}% full</option>`).join("");

document.getElementById("driverTasks").innerHTML=bins.filter(b=>b.fill>=75).map(b=>
`
 <div style="padding:12px 0;border-bottom:1px solid #eee"><b>${safe(b.name)}</b>
 <p class="muted">${safe(b.area)} · ${b.fill}% full</p><span class="tag warn">Collection required</span>
 <button class="secondary" onclick="chooseDriverBin('${b.id}')">Select</button></div>`).join("")||'<p class="notice">No urgent tasks. Check back later.</p>';
}
function chooseDriverBin(id){
 switchPage("driver");document.getElementById("driverBin").value=id;
}
```

```
function buildRoute(){
 const urgent=bins.filter(b=>b.fill>=75).sort((a,b)=>b.fill-a.fill);
 const out=document.getElementById("routeInfo");
 if(!urgent.length){out.textContent="No bins currently require urgent collection.";return}
 out.innerHTML="<b>Suggested priority route</b><ol>"+urgent.map(b=>
 `<li>${safe(b.name)} — ${b.fill}% full</li>`).join("")+"</ol><small>This demo prioritizes highest fill level first; it does not calculate road distances.</small>";
 if(map){map.fitBounds(urgent.map(b=>[b.lat,b.lng]),{padding:[25,25]})}
}
function collectBin(){
 const id=document.getElementById("driverBin").value;
 const b=bins.find(x=>x.id===id);
 const kg=Number(document.getElementById("loadKg").value);
 if(!b||!Number.isFinite(kg)||kg<=0){toast("Choose a bin and enter a valid collected load.");return}
 collectedKg+=kg;b.fill=0;b.status="Collected";b.collected=true;
 persist();log(document.getElementById("driverName").value+" collected "+kg+" kg at "+b.name);
 refresh();renderDriver();toast("Collection recorded: "+kg+" kg.");
}
document.getElementById("complaintForm").addEventListener("submit",e=>{
 e.preventDefault();
 const c={id:"CMP-"+Math.floor(100000+Math.random()*900000),
 name:document.getElementById("citizenName").value.trim(),
 email:document.getElementById("citizenEmail").value.trim(),
 type:document.getElementById("issueType").value,
 area:document.getElementById("issueArea").value.trim(),
 desc:document.getElementById("issueDesc").value.trim(),
 status:"Pending",date:new Date().toLocaleString()};
```

```
 complaints.unshift(c);persist();log("New public complaint "+c.id);
 e.target.reset();document.getElementById("trackId").value=c.id;
 trackComplaint();refresh();toast("Complaint registered: "+c.id);
});
function trackComplaint(){
 const id=document.getElementById("trackId").value.trim().toUpperCase();
 const c=complaints.find(x=>x.id===id);
 document.getElementById("trackResult").innerHTML=c?`
 <div class="notice"><b>${safe(c.id)}</b><p>${safe(c.type)} — ${safe(c.area)}</p>
 <p>${safe(c.desc)}</p><span class="tag
${c.status==="Resolved"?"":"warn"}">${safe(c.status)}</span>
 <p class="muted">Submitted: ${safe(c.date)}</p></div>`:'<p class="muted">Enter a
valid complaint ID to see its status.</p>';
}
function adminLogin(){
 if(document.getElementById("adminPass").value==="admin123"){
  adminUnlocked=true;document.getElementById("adminTools").style.display="block";
  renderAdmin();toast("Admin tools unlocked.");
 }else toast("Incorrect password.");
}
document.getElementById("binForm").addEventListener("submit",e=>{
 e.preventDefault();if(!adminUnlocked)return;
 const id="BIN-"+Math.floor(100+Math.random()*900);
 bins.push({id,name:document.getElementById("newBinName").value.trim(),
 area:document.getElementById("newBinArea").value.trim(),
 fill:Number(document.getElementById("newBinFill").value),
 capacity:Number(document.getElementById("newBinCapacity").value),
 lat:13.05+(Math.random()-.5)*.1,lng:80.23+(Math.random()-.5)*.1,
 status:"Normal",collected:false});
```

```
 persist();log("Administrator created
"+id);e.target.reset();refresh();renderAdmin();toast("Garbage bin added.");

});

function renderAdmin(){

 if(!adminUnlocked)return;


document.getElementById("adminBinTable").innerHTML=`<table><tr><th>Bin</th><th
>Fill</th><th>Status</th><th>Action</th></tr>`+

 bins.map(b=>`<tr><td><b>${safe(b.name)}</b><br>${safe(b.id)}</td>

 <td><select id="fill-${safe(b.id)}">${[0,10,25,50,75,90,100].map(n=>

 `<option value="${n}"
${b.fill===n?'selected':''}>${n}%</option>`).join("")}</select></td>

 <td>${safe(b.status)}</td><td><button onclick="updateBin('${b.id}')">Update</button>

 <button class="danger"
onclick="deleteBin('${b.id}')">Delete</button></td></tr>`).join("")+"</table>";

 document.getElementById("adminComplaintTable").innerHTML=complaints.length?

 `<table><tr><th>ID / Issue</th><th>Area</th><th>Status</th><th>Action</th></tr>`+


complaints.map(c=>`<tr><td><b>${safe(c.id)}</b><br>${safe(c.type)}<br>${safe(c.des
c)}</td>

 <td>${safe(c.area)}</td><td>${safe(c.status)}</td><td>

 <select id="cs-${c.id}">${["Pending","In Progress","Resolved"].map(s=>

 `<option ${c.status===s?'selected':''}>${s}</option>`).join("")}</select>

 <button
onclick="updateComplaint('${c.id}')">Save</button></td></tr>`).join("")+"</table>"

 :'<div class="empty">No complaints submitted.</div>';

}

function updateBin(id){

 const b=bins.find(x=>x.id===id);if(!b)return;

 b.fill=Number(document.getElementById("fill-"+id).value);
```

```
 b.status=b.fill>=75?"Needs collection":"Normal";
 b.collected=false;persist();log("Bin level updated: "+b.name);
 refresh();renderAdmin();toast("Bin status updated.");
}
function deleteBin(id){
 bins=bins.filter(b=>b.id!==id);persist();log("Bin deleted: "+id);
 refresh();renderAdmin();toast("Bin deleted.");
}
function updateComplaint(id){
 const c=complaints.find(x=>x.id===id);if(!c)return;
 c.status=document.getElementById("cs-"+id).value;persist();
 log("Complaint "+id+" marked "+c.status);refresh();renderAdmin();toast("Complaint status updated.");
}
function exportReport(){
 const rows=[["Bin ID","Name","Area","Fill %","Capacity kg","Status"],
 ...bins.map(b=>[b.id,b.name,b.area,b.fill,b.capacity,b.status])];
 rows.push([]);rows.push(["Complaint ID","Issue","Area","Status"]);
 complaints.forEach(c=>rows.push([c.id,c.type,c.area,c.status]));
 const csv=rows.map(row=>row.map(v=>'"'+String(v??"").replace(/"/g,'""')+'"').join(",")).join("\r\n");
 const a=document.createElement("a");a.href=URL.createObjectURL(new Blob([csv],{type:"text/csv"}));
 a.download="ecotrack-report.csv";a.click();URL.revokeObjectURL(a.href);
}
function resetDemo(){
 if(!confirm("Reset all saved EcoTrack demonstration data?"))return;
 bins=structuredClone(initialBins);complaints=[];activity=["Demo data reset"];
```

```
 collectedKg=125;persist();refresh();renderAdmin();toast("Demo data reset.");
}
function initChart(){
 if(!window.Chart)return;
 chart=new Chart(document.getElementById("wasteChart"),{
 type:"bar",
 data:{labels:["Mon","Tue","Wed","Thu","Fri","Sat","Sun"],
 datasets:[{label:"Waste collected (kg)",data:[320,410,350,490,430,560,380],
 backgroundColor:"#36a873",borderRadius:7}]},
 options:{responsive:true,maintainAspectRatio:false,
 plugins:{legend:{display:false}},
 scales:{y:{beginAtZero:true,grid:{color:"#edf1ed"}},x:{grid:{display:false}}}}
 });
}
function refresh(){

updateStats();renderBins();renderMarkers();renderPriority();renderActivity();renderDrive
r();
 if(adminUnlocked)renderAdmin();
}
initMap();initChart();refresh();

/* Simulated live updates for demonstration; not connected to real sensors. */
setInterval(()=>{
 bins.forEach(b=>{
  if(!b.collected&&b.fill<100)b.fill=Math.min(100,b.fill+Math.floor(Math.random()*3));
  b.status=b.fill>=75?"Needs collection":"Normal";
 });
 persist();refresh();
```

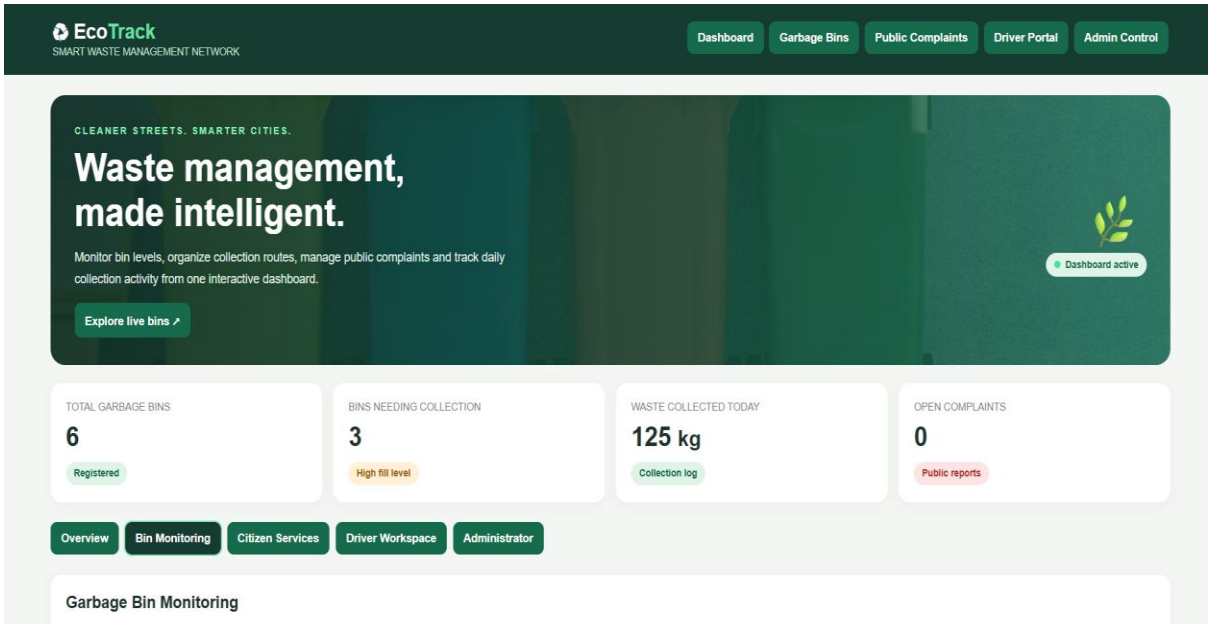
},10000);

</script>

</body>

</html>

Output:



Result: Thus, the Smart Waste Management System was implemented using HTML, CSS, JavaScript, interactive maps, and charts.